

DIPLOMA EXAMINATION IN ENGINEERING/ TECHNOLOGY

Model question paper

POWER PLANT INSTRUMENTATION

Time:3Hours

(Maximum Marks:100)

PART -A

MARKS:10

(Answer the following in one or two sentences.Each question carries 2 marks each)

I.

1. Define alternator efficiency
2. Define Nuclear fussion
3. Define priming
4. List types of Furnace draft control
5. Write down the electrical parameters measured in turbine system (5x2=10)

PART B

(Maximum Marks:30)

(Answer any five questions from the following .Each question carries 6 marks)

II.

1. Compare conventional and non conventional energy sources
2. Explain the feed water circuit
3. Explain steam pressure control of boiler
4. Illustrate five element drum level control
5. list the components of waste gas
6. Explain the types of fuel used in boiler
7. List the safety parameters measured in turbine and explain any two (5x6=30)

PART C

(Maximum Marks:60)

(Answer one full question from each module. Each question carries 15 marks)

MODULE I

- III. a). Draw the lay out of thermal power plant and Explain it 15

OR

- IV a) List the main components of nuclear reactor and explain it 9

- b) Explain the cooling water circuit 6

MODULE II

- V a) Explain the measurements in feedwater circuit 9
b) Describe the measurements of impurities of water 6

OR

- VI a) Explain the sequencing of burner and other operations 7
b) Describe the effect of impurities of water 8

MODULE III

- VII a) Illustrate the working of combustibles analyzer 8
b) Explain the combustion control of liquid fuel 7

OR

- VIII . a) Describe the metered cross limited air fuel ratio control 8
b) Explain induced draft control 7

MODULE IV

- IX a) Draw the diagram of turbine lubrication system and explain it 10
b) Explain the master control 5

OR

- X a) List the measurements in turbine process control system 6
b) Explain the classification of turbine 9